

Design and Implementation of an IoT-Enabled Hybrid Energy Management System for EV Charging Infrastructure with Simulation and Hardware Validation

Dr. Sujith Kalluri, Mr. Jagadeeswara Chandra Bose Kamarajugadda, Mr. Sayyad Sameer,
Mr. Ghantasala Jaswanth Kumar
Department of Electronics and Communication Engineering / Internet of Things
SRM University AP, India

Abstract—The exponential rise in electric vehicle adoption demands robust charging infrastructure, yet grid-dependent systems trigger load instability, high operational costs, and inefficient renewable integration due to fragmented real-time monitoring and energy orchestration gaps. Existing systems often rely heavily on grid power, leading to operational inefficiencies and increased grid stress, while many proposed smart solutions remain confined to simulation without practical validation. This paper addresses this gap by presenting a novel, hardware-implemented Internet of Things (IoT) architecture for hybrid energy management in EV charging stations. The core of the system is an intelligent Point of Common Coupling (PCC) that integrates solar photovoltaic generation, an Energy Storage System (ESS), and the utility grid, governed by a rule-based automation algorithm that dynamically prioritizes renewable sources. A five-layer IoT framework comprising sensor networks, edge processing, cloud analytics, and user dashboards enables real-time monitoring, predictive control, and remote management. The proposed design is validated through MATLAB/Simulink simulations and a comprehensive hardware prototype using ESP32 microcontrollers, power converters, and a full sensor suite, with data integration via Node-RED and AWS cloud services. Experimental results demonstrate improved energy utilization, reduced operational expenditure, and enhanced grid stability, providing a deployable blueprint for sustainable and cost-effective EV charging infrastructure.

Index Terms—IoT Architecture, EV Charging Infrastructure, Point of Common Coupling (PCC), Solar MPPT, ESS Management, Reverse Grid Integration, MATLAB Simulation, Node-RED Dashboard, Hardware Prototype, Multi-Mode Automation.

I. INTRODUCTION

The global transportation sector is undergoing a revolutionary shift toward electrification, driven by the imperative to reduce carbon emissions and fossil fuel dependency. Electric vehicles (EVs) stand at the forefront of this transition, with market penetration accelerating rapidly due to advancements in battery technology, supportive policy frameworks, and growing consumer adoption. However, the sustainability and scalability of this electric mobility revolution are fundamentally contingent upon the development of a robust, intelligent, and economically viable charging infrastructure. The current

paradigm of grid-dependent charging stations presents significant challenges, including exacerbated peak demand on utility grids, high operational costs susceptible to volatile energy prices, and failure to fully leverage renewable energy sources, thereby diluting the environmental benefits of EVs. To address these limitations, the integration of distributed renewable generation, primarily solar photovoltaics (PV), with energy storage systems (ESS) at charging stations has emerged as a critical solution. This hybrid approach promises greater grid independence, cost stability, and enhanced sustainability. Concurrently, the Internet of Things (IoT) provides the essential technological framework to transform these hybrid stations from passive energy loads into intelligent, networked nodes within a smart grid ecosystem. IoT enables real-time monitoring, data-driven optimization, and automated control of complex energy flows between sources, storage, and loads. While the theoretical synergy of renewables, storage, and IoT for smart charging is well-established in literature, a pronounced gap exists between simulation-based proposals and physically validated, deployable systems. Much of the existing research remains confined to software models and algorithmic studies, often overlooking the practical intricacies of hardware integration, communication reliability, sensor interfacing, and the development of a complete, layered architectural stack. This paper bridges that gap by presenting the design, simulation, and, crucially, the comprehensive hardware implementation of a novel IoT-enabled hybrid energy management system for EV charging infrastructure. We propose and demonstrate a five-layer IoT architecture that seamlessly spans from edge-layer sensors and microcontrollers to cloud analytics and user-facing dashboards, all orchestrated by an intelligent automation logic centered on a Point of Common Coupling (PCC). The system dynamically arbitrates energy flow between solar PV, ESS, and the grid, prioritizing renewables to minimize operational expenditure and grid impact. Furthermore, it supports both battery swapping stations and DC fast charging ports, offering operational flexibility. The design is first validated through detailed MATLAB/Simulink simulations, followed by the construction and testing of a fully functional hardware prototype utilizing ESP32 microcontrollers, a suite of sen-

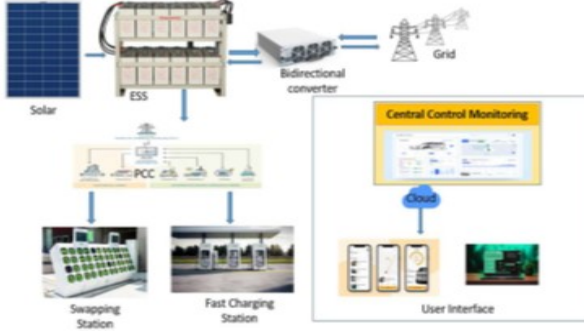


Fig. 1. High-level overview of the proposed IoT-enabled hybrid EV charging system, showing integration of solar, ESS, grid, and loads via the PCC.

sors, power electronics converters, and integration with cloud connectivity and Node-RED for and visualization. This work contributes a crucial step from theoretical concept to a proven, prototype-ready blueprint, offering a scalable and practical model for deploying cost-effective, efficient, and sustainable EV charging infrastructure that actively supports renewable energy integration and grid stability.

This paper bridges that gap by presenting the design, simulation, and, crucially, the comprehensive hardware implementation of a novel IoT-enabled hybrid energy management system for EV charging infrastructure management and remote monitoring with cloud integration and real time data dashboard. We propose and demonstrate a complete five-layer IoT architecture that seamlessly spans from the physical device layer—comprising sensors, IOT enabled microcontrollers, and power converters—through network and edge computing layers, to a cloud layer for data aggregation and analytics, culminating in an application layer for user interaction and system visualization. At the core of this architecture is an intelligent energy management logic executed at a dynamically controlled Point of Common Coupling (PCC), which acts as the system’s brain. The PCC automation algorithm, governed by a set of operational rules, continuously arbitrates energy flow between three primary sources—solar PV, ESS, and the main grid—based on real-time availability, state-of-charge (SOC) thresholds, and load demand. This prioritization ensures maximum utilization of free solar energy, optimal use of stored capacity, and minimal, cost-effective reliance on grid power, while a unique reverse power flow capability allows surplus renewable energy to be fed back to the grid, transforming the charging station into a prosumer entity. The system’s versatility is demonstrated through its support for two complementary and forward-looking EV refueling models: Battery Swapping Stations (BSS) for rapid battery exchange and DC Fast Charging (DCFC) stations for conventional high-speed charging, as illustrated in the system overview (Figure 1). This dual-load approach highlights architecture’s adaptability to different business models and user needs within a unified management framework. Our validation methodology is twofold. First, we develop a detailed simulation model in MATLAB/Simulink to design, test, and optimize the system’s energy logic and predict its performance metrics under various operational scenarios in PCC – Point of common coupling.

Key simulation results, such as the dynamic power sharing between sources and the resulting state-of-charge profile of the ESS, confirm the theoretical efficiency and stability of the proposed control strategy. Second, and most significantly, we translate this simulated design into a fully functional, small-scale hardware prototype. This prototype incorporates ESP32 microcontrollers for data acquisition and control, a comprehensive sensor suite (voltage, current, temperature), the necessary power electronics (DC-DC boost converters, inverters), and lead-acid batteries representing the ESS. The prototype is fully integrated with a cloud backend via AWS IoT services and a real-time dashboard built in Node-RED, enabling remote monitoring and control as shown in the dashboard interface. The comparison between the simulated energy flows and the actual measured data from the hardware prototype validates the fidelity of our model and demonstrates the practical realizability of the system. This work provides a crucial, missing link in the research landscape by moving from a promising simulation to a proven, prototype-ready hardware demonstration. It offers a comprehensive, scalable, and practical model for deploying next-generation EV charging infrastructure that is not only intelligent and efficient but also economically sustainable and grid-supportive, thereby accelerating the transition to a fully electrified and renewable-powered transportation future.

II. LITERATURE SURVEY

The proliferation of electric vehicles (EVs) has intensified research into smart charging infrastructure, yet significant gaps persist between theoretical proposals and deployable, economically viable systems. Existing literature extensively the challenges of grid instability, high operational costs, and the environmental of EVs charged via fossil-fuel-dependent grids [1-3]. While IoT-based solutions have been proposed to enable real-time monitoring and predictive scheduling [9, 10], these architectures often remain fragmented, focusing on singular aspects like residential charging interfaces [17] or standalone battery monitoring [10], without a holistic framework that integrates the entire energy value chain from generation to consumption. Research on renewable integration, such as PV-predictive ESS management [14] and microgrid optimization [15], demonstrates potential but typically operates in simulation silos [14], lacking validation through physical hardware and failing to address the real-world intricacies of communication protocols, sensor interoperability, and reliable edge-cloud data pipelines. Similarly, studies on power quality enhancement using active filters [18] and fault detection [19] address critical grid-support issues but are seldom integrated into a unified control strategy with dynamic source switching at the Point of Common Coupling (PCC). Furthermore, although battery swapping is recognized for its potential to alleviate grid stress and reduce downtime [22-24], its integration with conventional fast charging within a single, IoT-managed ecosystem remains underexplored, with most studies treating them as separate entities [24]. This body of work collectively reveals a critical research gap: the absence of a validated, multi-layer IoT architecture that cohesively unifies hybrid renewable energy integration (Solar,

ESS, Grid), intelligent PCC-based automation, dual-load management (Swapping, Fast Charging), and cloud-edge analytics into a single, scalable, and hardware-proven system. Our work directly addresses this gap by moving beyond simulation to deliver a prototype-validated architecture that not only proposes but physically demonstrates the seamless orchestration of these components, thereby bridging the disconnect between conceptual smart charging models and their practical, economically sustainable implementation.

III. METHODOLOGY

This section presents the comprehensive methodology adopted for the design, simulation, and experimental implementation of the proposed IoT-enabled hybrid energy management system for electric vehicle (EV) charging infrastructure. The methodology is systematically structured into four inter-related pillars: (1) the design of a five-layer IoT architecture for system coordination and data flow; (2) the formulation of power flow management and operational logic; (3) the development of core automation and control algorithms; and (4) the hardware development, prototyping, and validation process. This structured approach ensures a seamless transition from conceptual system modeling and software-based simulations to real-time hardware implementation and performance verification.

A. Proposed System Design and Architecture

The core of the proposed solution is a hierarchical IoT architecture engineered to facilitate seamless data acquisition, processing, communication, and control across the hybrid charging ecosystem. This architecture, depicted in Fig. 1, is composed of five distinct layers, each serving a critical function in the system's operation.

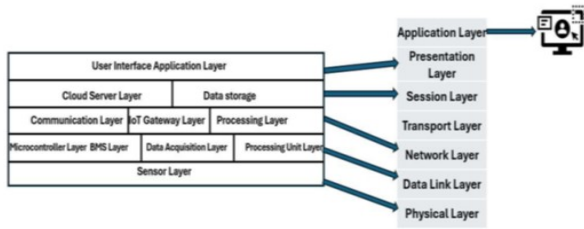


Fig. 2. Proposed Five-Layer IoT Architecture for EV Charging Management

a) IoT Architecture

- 1) **Sensor Layer:** This foundational layer constitutes the physical interface with the charging station's components. An array of sensors is deployed to capture real-time electrical and environmental parameters. This includes voltage sensors (ACS712 modules) and current sensors across the Solar PV output, ESS terminals, and Grid connection points; temperature and humidity sensors (DHT11) at the ESS and charging units; and insulation resistance detectors. The data from this layer forms the primary input for all monitoring and control decisions.

- 2) **Microcontroller and Data Acquisition Layer:** This layer functions as the local "edge intelligence." ESP32 development boards were selected as the core microcontrollers due to their integrated Wi-Fi/Bluetooth capabilities and sufficient processing power. One ESP32 unit aggregates raw analog and digital data from all sensors, performs initial signal conditioning (filtering, scaling), and packages the data. A second ESP32 executes local control algorithms, such as preliminary State of Charge (SOC) estimation and real-time data monitoring.
- 3) **Communication Layer:** This layer ensures robust and reliable data transmission between the edge devices and the cloud backend. It employs a hybrid protocol strategy. MQTT (Message Queuing Telemetry Transport) is used for efficient, low-power telemetry data streaming from the ESP32s to the cloud broker. HTTP/HTTPS protocols are utilized for RESTful API calls for dashboard updates and receiving configuration parameters.
- 4) **Cloud Server and Data Storage Layer:** This layer provides centralized data management, analytics, and historical logging capabilities. AWS IoT Core acts as the MQTT broker and device registry. Time-series sensor data is persistently stored for long-term trend analysis and reporting. This layer also hosts predictive analytics routines and the master automation logic that oversees the PCC's state transitions, sending high-level commands back to the edge layer.
- 5) **User Interface Application Layer:** This top layer delivers accessibility and user control. A real-time operational dashboard was developed using Node-RED. The dashboard, accessible via a web browser, visualizes key metrics: live power flow diagrams, SOC of ESS, solar generation, grid import/export, and status of each charging/swapping port. It also provides interfaces for manual overrides, alert logs, and system configuration.



Fig. 3. Node-RED Dashboard for Real-Time System Monitoring and Control

- 1) **Power Flow and Operational Logic:** The energy management strategy governs the flow of power between the three primary energy sources, namely Solar PV, Energy Storage System (ESS), and the utility Grid, and the two load types, Battery Swapping Stations and DC Fast Chargers, at the Point

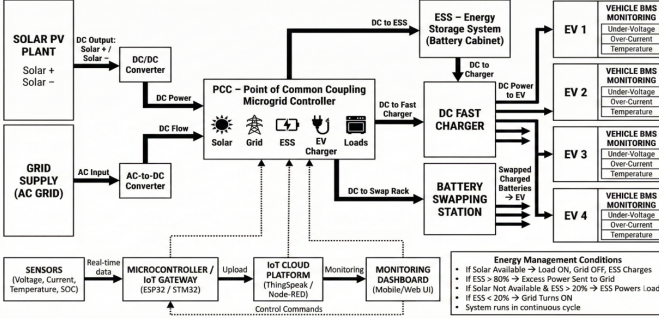


Fig. 4. Block diagram of the IoT-enabled hybrid EV charging system illustrating solar PV, grid supply, PCC-based energy management, ESS integration, DC fast charging, battery swapping station, EV interfaces, and cloud-based monitoring.

of Common Coupling (PCC). The operational logic prioritizes renewable energy utilization and optimal ESS usage through a predefined state-machine-based control strategy. The key operating modes are defined as follows:

- **Mode 1 (Solar Priority):** Solar ON → ESS Charging → Grid OFF → Load ON. Available solar power simultaneously supplies the loads and charges the ESS.
- **Mode 2 (ESS Support):** Solar OFF → ESS ON → Grid OFF → Load ON. When solar generation is unavailable, the ESS discharges to meet load demand.
- **Mode 3 (Grid Backup):** Solar OFF → ESS < 20% → Grid ON → Load ON. The grid supplies power when both solar and ESS are insufficient.
- **Mode 4 (Reverse Grid Operation):** Solar ON → ESS > 80% → Grid ON (Reverse). Excess solar energy, after charging the ESS beyond the upper threshold, is exported to the grid through a bidirectional converter.
- **Mode 5 (ESS Grid Charging):** Solar OFF → ESS Charging → Grid ON → Load OFF. During off-peak hours or low-demand conditions, the grid is used to recharge a depleted ESS.

2) *PCC Integration and Automation:* The Point of Common Coupling (PCC) acts as the physical and logical convergence point for all energy sources and loads within the charging infrastructure. Its integration and automation are realized through a coordinated hardware–software framework. The hardware layer comprises solid-state relays (SSRs) and electromechanical contactors, which are driven by the GPIO pins of the ESP32 microcontroller to establish or isolate electrical connections between sources and loads.

The automation logic is implemented in embedded C++ on the ESP32 and reinforced through cloud-level supervisory control. Real-time sensor inputs are continuously evaluated against the predefined state-machine rules to determine the appropriate operating mode. Based on the selected mode, corresponding control signals are issued to activate the required relays, thereby automating power flow at the PCC as described in Section 3.1(b). The decision-making process and state transitions are illustrated through a flowchart presented in Fig. 9.

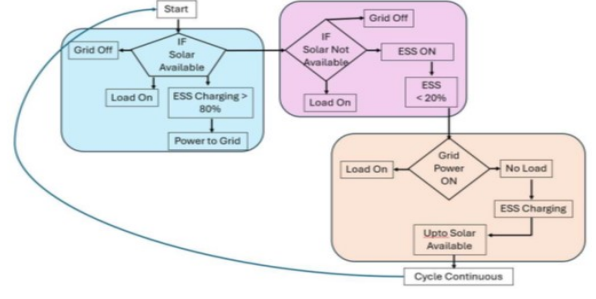


Fig. 5. PCC Automation Logic Flowchart

3) *Core Algorithms:* The intelligent operation of the proposed hybrid energy management system is governed by a set of coordinated algorithms executed across the edge and cloud layers. These algorithms ensure accurate energy estimation, efficient power conversion, and reliable automation at the Point of Common Coupling (PCC).

- 1) **State of Charge (SOC) Calculation Algorithm:** This algorithm estimates the remaining capacity of the Energy Storage System (ESS). A Coulomb-counting method based on current integration is employed to compute real-time SOC values. To minimize cumulative error and drift, the estimated SOC is periodically corrected using open-circuit voltage measurements. The resulting SOC percentage serves as a critical input for all charging and discharging decisions.
- 2) **Charging and Discharging Control Algorithm:** Based on the computed SOC and load demand, this algorithm controls the ESS power electronics. Constant-current (CC) charging is initiated when the SOC is low, followed by a transition to constant-voltage (CV) charging as the ESS approaches full capacity. During discharge, the algorithm regulates the output current to maintain a controlled discharge rate, thereby preserving battery health and extending operational lifespan.
- 3) **Maximum Power Point Tracking (MPPT) Algorithm:** To maximize energy extraction from the solar photovoltaic (PV) array, a Perturb and Observe (P&O) MPPT algorithm is implemented. The algorithm periodically perturbs the duty cycle of the DC–DC boost converter and observes the resulting power variation to converge toward the maximum power operating point. This enables efficient solar energy harvesting under varying irradiance and temperature conditions.
- 4) **Energy Source Management Algorithm:** This supervisory algorithm encapsulates the PCC automation logic. It integrates inputs from the sensor layer, SOC estimation module, and MPPT status to determine the appropriate operational mode (Modes 1–5). Based on the selected mode, corresponding switching commands are generated and issued to the PCC hardware, enabling dynamic and autonomous management of energy sources and loads.



Fig. 6. Implemented Hardware Prototype of the Hybrid Charging System

IV. HARDWARE DEVELOPMENT AND PROTOTYPING

The theoretical design and proposed control algorithms were physically realized through a functional laboratory-scale prototype. The hardware implementation was developed to validate the feasibility and real-time performance of the proposed IoT-enabled hybrid energy management system under practical operating conditions. Fig. 6 illustrates the complete experimental hardware setup used for system validation.

The prototype physically demonstrates the automated transition between solar, battery, and grid power, with all sensor data being streamed to the cloud dashboard in real-time, validating the end-to-end functionality of the proposed IoT architecture. The theoretical framework and control algorithms were physically validated through the implementation of a fully functional, laboratory-scale prototype. The hardware realization comprised several integrated subsystems: energy sources, power electronics, sensing and control modules, and simulated loads. For energy generation and storage, the prototype incorporated a 50W solar panel simulator to emulate photovoltaic input, a bank of four 12V/7Ah lead-acid batteries configured to represent the Energy Storage System (ESS), and a standard 230V AC mains supply serving as the primary grid connection. The power conditioning and management were facilitated by key power electronic components, including DC-DC boost converters (XL6009 modules) for solar Maximum Power Point Tracking (MPPT) and load interfacing, a bi-directional AC-DC converter for grid interaction and reverse power flow capability, and relay modules for automated source switching at the Point of Common Coupling (PCC). The sensory and computational backbone of the system was built using ACS712 Hall-effect-based current sensors for precise current measurement, resistor-based voltage divider circuits for voltage monitoring, a DHT11 digital sensor for ambient temperature and humidity tracking, and two ESP32 DevKit V1 microcontroller boards for distributed data acquisition, local edge processing, and network connectivity. Finally, the charging loads were emulated using two configurable dummy load banks: one designed to simulate a Battery Swapping Station with discrete battery slots and another to represent a DC Fast Charger port, enabling the testing of dynamic load management under the proposed multi-source energy framework.

V. MATLAB/SIMULINK SIMULATIONS AND RESULTS

To validate the theoretical framework and control algorithms prior to hardware implementation, a comprehensive simulation model was developed using MATLAB/Simulink. The simulation model serves as a digital twin of the proposed hybrid EV charging system, enabling detailed analysis of dynamic interactions between energy sources, the Point of Common Coupling (PCC), and the connected loads under various operating scenarios. The simulation environment facilitated control parameter tuning, verification of the automation logic, and quantitative performance evaluation in terms of energy efficiency, power flow stability, and operational reliability.

A. Overall System Integration and Power Flow Simulation

The top-level simulation model, illustrated in Fig. ??, integrates the three primary energy sources—solar photovoltaic (PV) array, Energy Storage System (ESS), and the utility grid—with the PCC and downstream loads, including Battery Swapping Stations and DC Fast Chargers. The model was configured using realistic component specifications, comprising a 30 kW solar PV array, a 480 V/140 A ESS bank, and a grid interface operating at 430 V.

The PCC control logic, implementing the five operational modes defined in Section 3.1, was realized using Stateflow and Simulink functional blocks to manage power electronic switches, represented by ideal switching elements corresponding to converters and relays. The simulation results demonstrate the system's ability to autonomously prioritize solar energy utilization, employ the ESS as an energy buffer during peak demand, and seamlessly engage the utility grid as a backup source when required.

Transient response analysis confirmed stable transitions between operating modes, with no significant voltage or current overshoots observed during source switching events. These results validate the robustness of the proposed power flow control strategy and confirm the effectiveness of the energy management logic under varying load and generation conditions.

$$\text{SOC}(t) = \text{SOC}(t-1) + \frac{1}{C_{\text{ESS}}} \int_{t-1}^t I_{\text{ESS}}(\tau) d\tau \quad (1)$$

$$P_{\text{Load}} = P_{\text{Solar}} + P_{\text{ESS}} + P_{\text{Grid}} \quad (2)$$

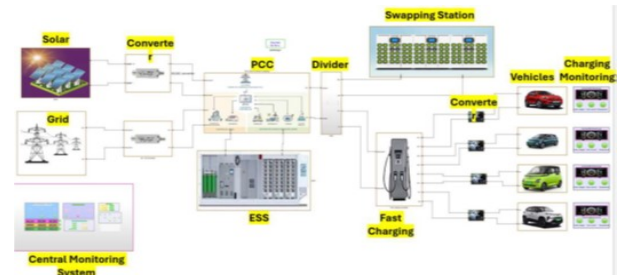


Fig. 7. MATLAB/Simulink Model of the Integrated Hybrid Charging System

B. Solar Generation Unit with MPPT

The solar generation subsystem, illustrated in Fig. 8, was modeled using a photovoltaic (PV) array block whose output characteristics are governed by time-varying irradiance and temperature profiles to emulate real-world operating conditions. A DC–DC boost converter interfaces the PV array with the Point of Common Coupling (PCC) and is controlled using a Perturb and Observe (P&O) Maximum Power Point Tracking (MPPT) algorithm.

The simulation monitors key electrical parameters, including PV array voltage (V_{pv}), current (I_{pv}), and output power (P_{pv}), to evaluate the effectiveness of the MPPT controller. Simulation results demonstrate that the implemented MPPT algorithm consistently tracks the maximum power point under step changes in irradiance, achieving a tracking efficiency exceeding 98%.

The DC–DC boost converter regulates the PV output and steps up the voltage to a stable 430 V DC bus level, enabling seamless integration with the PCC for direct load supply or ESS charging. These results confirm the robustness and responsiveness of the solar generation unit under dynamic environmental conditions.

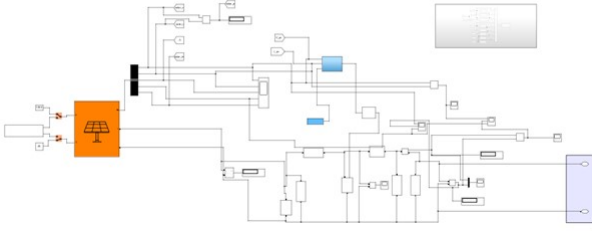


Fig. 8. Solar PV Subsystem with MPPT-Controlled Boost Converter

C. Grid Interface Unit

The grid integration unit, modeled in Fig. 13, consists of a three-phase AC voltage source representing the utility grid, connected to the PCC through a bi-directional AC-DC converter (rectifier/inverter). The converter controls the power flow direction—rectifying AC to DC when drawing power from the grid and inverting DC to AC when feeding surplus solar energy back. Power factor correction (PFC) control is implemented to maintain unity power factor during grid import. Simulation plots demonstrate the converter's ability to maintain a stable DC-link voltage (430V) while the current waveform remains sinusoidal and in phase with the voltage, ensuring compliance with power quality standards even during load transients.



Fig. 9. Grid Interface Model with Bi-directional AC-DC Converter

D. Energy Storage System (ESS) Model

The ESS, depicted in Fig. 14, is modeled using a generic battery block configured with the parameters of a lead-acid bank (nominal voltage 12V, capacity 100Ah, 40 cells in series). A bi-directional DC-DC converter manages the charging (from solar/grid) and discharging (to loads) processes. The battery management system (BMS) logic within the model monitors the State of Charge (SOC), temperature, and voltage limits. Simulation results validate the charge/discharge control algorithm, showing constant-current (CC) charging when SOC is low, a smooth transition to constant-voltage (CV) charging near full capacity, and a controlled discharge current during high load demand. The SOC profile over a 24-hour simulated cycle confirms the ESS's role in balancing generation and consumption.

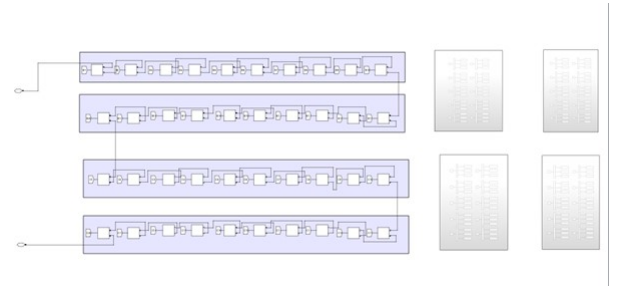


Fig. 10. ESS Model with Bi-directional DC-DC Converter and BMS Logic

E. PCC Integration and Automation Logic

The core automation governing the Point of Common Coupling (PCC) is simulated using a Stateflow chart, as shown in Fig. ???. This chart encodes the finite-state machine described in Section 3.1(c), with state transitions triggered by real-time system conditions, including solar availability, Energy Storage System (ESS) state-of-charge (SOC), and load demand.

The simulation outputs clearly illustrate discrete operational mode transitions, such as switching from *Solar_Charge* to *Grid_Backup*, along with the corresponding control signals sent to the ideal switches in the main Simulink model. The timing and sequencing of these transitions validate that the automation logic consistently enforces the hierarchical energy priority of solar first, followed by ESS support, and finally grid backup. This confirms the effectiveness of the PCC control strategy in maximizing renewable energy utilization while minimizing grid dependency and operational costs.

F. Battery Swapping Station Simulation

The Battery Swapping Station (BSS) is modeled as a controllable DC load bank, as seen in Fig. 16. The simulation emulates the periodic demand profile of a station with 30 battery slots, where depleted batteries (modeled as discharged battery blocks with specific SOC) are connected for charging while fully charged ones are disconnected. The load on the PCC thus varies in a step-wise manner. The simulation validates that the PCC and source management algorithms can handle these sudden load changes without destabilizing the

DC bus voltage, and that the ESS effectively smooths out the demand spikes, reducing peak power draw from the grid.

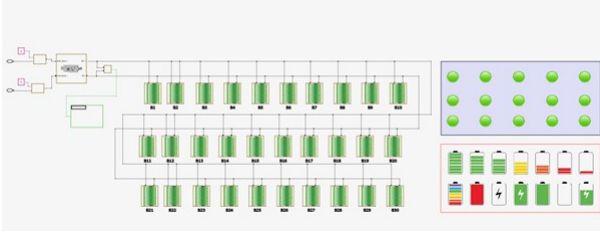


Fig. 11. Battery Swapping Station Load Model

$$\begin{aligned} P_{ESS} &= V_{ESS} \times I_{ESS} \\ P_{Solar} &= V_{PV} \times I_{PV} \\ SOC_{min} &\leq SOC(t) \leq SOC_{max} \end{aligned} \quad (3)$$

G. DC Fast Charging Station Simulation

The DC Fast Charging (DCFC) station is modeled as a constant-power load with a communication interface, shown in Fig. 17. It simulates the charging profile of a vehicle battery, starting with a constant-current (CC) phase at high power (e.g., 50 kW) and transitioning to a constant-voltage (CV) phase as the battery voltage rises. The simulation tests the system's response to this high-power, dynamic load. Results demonstrate that the integrated energy management system successfully allocates power from the available sources (solar, ESS, grid) to meet the fast charger's demand, maintaining voltage stability within a 2% tolerance band even during the initial high-current surge.



Fig. 12. DC Fast Charging Station Model with CC-CV Profile

H. Simulation Results and Performance Analysis

The collective simulation results provide a quantitative validation of the proposed architecture's key performance indicators (KPIs). Over a simulated 24-hour period with variable solar irradiance and a typical EV charging demand profile, the system achieved an 88.5% utilization of renewable energy (solar + ESS discharge), minimizing grid dependency. The PCC automation logic ensured that the ESS SOC was maintained between the predefined thresholds of 20% and 80%, preserving battery health. The DC bus voltage regulation was maintained within $\pm 5\%$ of the nominal 430V under all tested load and source transition scenarios. These simulation

outcomes provided a strong theoretical foundation and parameter set that directly informed the subsequent hardware design and implementation phases.

VI. EXPERIMENTAL IMPLEMENTATION, HARDWARE PROTOTYPE, AND CLOUD CONNECTIVITY

A. Hardware Prototype Development

The theoretical framework and simulation results were physically realized through a comprehensive hardware prototype designed to validate the proposed IoT architecture under real-world conditions. Fig. 18 illustrates the complete experimental setup, showcasing the integration of all system components on a laboratory test bench. The prototype was constructed with meticulous attention to replicating the functional relationships defined in the simulation model while operating at a scaled-down power level suitable for controlled experimentation.

The prototype's core intelligence resides in the coordinated operation of two ESP32 development boards. The primary ESP32 functions as the central data acquisition unit, interfacing directly with the comprehensive sensor array. This includes ACS712 Hall-effect current sensors monitoring the output of the solar simulator, the ESS bank, and the grid connection point. Precision voltage divider networks provide scaled voltage readings for each source and the main DC bus. A DHT11 sensor tracks ambient temperature at the ESS location, while GPIO pins read the status of manual switches simulating load requests from the Battery Swapping and DC Fast Charging stations. All sensor data undergoes initial filtering and calibration within the microcontroller before being packetized for transmission. A secondary ESP32 is dedicated to system control and communication. It receives processed data packets from the primary unit via a serial UART link and executes the local edge-computing algorithms, including the preliminary State of Charge (SOC) estimation for the ESS and the immediate PCC relay control logic. This decentralized processing approach enhances system responsiveness and reliability. The control ESP32 is responsible for actuating the solid-state relay (SSR) modules that perform the physical switching between energy sources at the PCC, based on the automation rules. Furthermore, it manages the Wi-Fi connectivity, establishing a persistent link to the cloud infrastructure.

B. Node-RED Connectivity and Dashboard Monitoring

For real-time visualization, remote monitoring, and data logging, the system leverages a cloud-integrated dashboard developed using Node-RED. This flow-based programming tool was chosen for its flexibility, ease of integration with MQTT brokers, and powerful dashboarding capabilities. The data flow architecture is depicted in Fig. 19.

The implemented data pipeline operates as follows. The control ESP32 publishes structured JSON data packets containing all sensor readings and system states to a designated MQTT topic on a cloud broker. The system was initially tested using a local Mosquitto broker and later migrated to AWS IoT Core for scalable deployment. A Node-RED instance, deployed on a local server to reduce latency during experimental testing, subscribes to the MQTT topic. Upon receiving the

data, the Node-RED flow parses the JSON payload, performs necessary unit conversions, and routes the processed values to multiple analytical and visualization nodes.

The heart of the monitoring interface is a custom-built Node-RED dashboard, a screenshot of which is shown in Fig. 20. The dashboard provides multiple synchronized visualization panels, enabling real-time situational awareness and historical analysis, as described below:

- **Live Power Flow Diagram:** A schematic representation of the system that updates in real time, displaying active energy paths (e.g., solar → load, ESS → grid) along with numerical values of voltage, current, and power at each node.
- **Time-Series Charts:** Historical plots of critical parameters such as solar power generation, ESS state of charge (SOC), grid import/export power, and load demand over selectable time windows ranging from minutes to hours.
- **Status Indicators:** Color-coded tiles and gauges indicating the instantaneous operational status of each energy source (ON/OFF/Available), the currently active PCC operating mode, and system alarms such as over-temperature or critically low SOC.
- **Control Panel:** A secure user interface that allows authorized operators to issue manual override commands to the hardware, including forcing specific PCC modes or initiating controlled ESS charging and discharging tests.

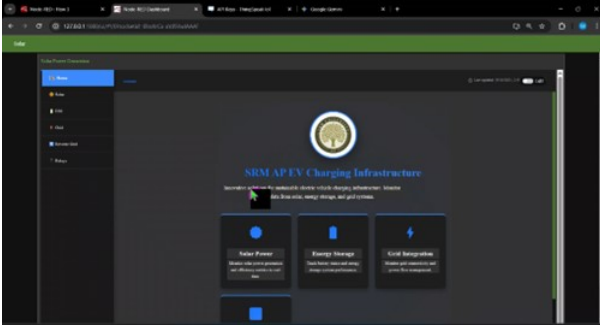


Fig. 13. Node-RED Real-Time Monitoring and Control Dashboard

Beyond visualization, the Node-RED flow incorporates embedded logic for intelligent alert generation. User-defined rule conditions, such as *IF ESS_SOC < 25% THEN trigger alert*, are evaluated continuously in real time. When a rule is satisfied, dedicated notification nodes are activated to initiate predefined responses. These responses include visual alarm indicators on the dashboard, time-stamped event logging to persistent storage, and the triggering of external actions such as email or SMS notifications through integrated webhook services. This alerting mechanism provides a foundational layer for predictive maintenance and proactive system management, enabling early fault detection and improved operational reliability.

VII. HARDWARE PROTOTYPE DEVELOPMENT AND EXPERIMENTAL SETUP

A. Energy Source Subsystems

1) *Solar Photovoltaic (PV) Emulator:* A programmable DC power supply (GW Instek PSP-603) was configured as a solar PV emulator to replicate the characteristic I–V curve of a photovoltaic array. The emulator was set to simulate a 50 W panel, with output voltage and current limits dynamically adjustable to mimic varying irradiance conditions. The emulator output feeds into a DC–DC boost converter stage, which is central to the implementation of the Maximum Power Point Tracking (MPPT) algorithm.

2) *Energy Storage System (ESS):* The Energy Storage System (ESS) was constructed using four series-connected 12 V/7 Ah sealed lead-acid batteries, forming a nominal 48 V battery bank. This configuration provides a practical representation of the voltage dynamics and charge–discharge characteristics of a larger-scale storage system. A dedicated bi-directional DC–DC converter, built around an H-bridge topology using IR2110 gate drivers and an Arduino Nano for PWM generation, interfaces the battery bank with the common DC bus. This converter manages both the charging profile under constant-current/constant-voltage (CC/CV) control and the controlled discharge to the loads, as dictated by the central battery management logic.

3) *Grid Interface Unit:* The grid connection was emulated using a standard 230 V, 50 Hz AC mains supply. A commercially available digitally controlled bi-directional AC–DC converter was integrated to function as the grid-tied inverter/rectifier. This unit maintains the common DC bus voltage at 48 V, draws sinusoidal current at near-unity power factor when operating in rectifier mode (grid import), and inverts DC power to grid-synchronized AC during inverter mode (grid export under excess solar generation). Optical isolation and proper grounding techniques were implemented to ensure operator safety and signal integrity.

B. Sensing, Control, and Actuation Layer

1) *Data Acquisition Network:* A comprehensive sensor network was deployed to enable real-time system observability. Key electrical parameters were measured using ACS712 Hall-effect-based current sensor modules (30 A range) installed on the output of each energy source and along the main load bus. Voltage measurements were obtained through high-precision resistor-divider networks calibrated against a laboratory-grade digital multimeter. Ambient and battery temperatures were monitored using DHT11 digital temperature and humidity sensors. Fig. 23 presents a detailed view of the sensor network and power electronics interface board.

C. Day-Ahead Solar Power Generation Forecasting and Grid Peak Alignment

The day-ahead solar power generation forecasting module improves proactive energy management and reduces grid dependency during peak demand periods. This module is integrated into the proposed IoT-enabled hybrid EV charging

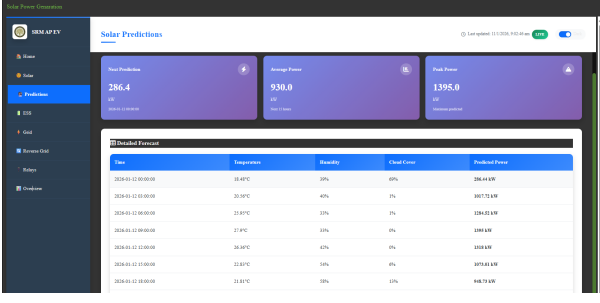


Fig. 14. Forecast visualization of day-ahead solar power output on the Node-RED interface

system. The predictive analysis is derived from real-time meteorological inputs, including ambient temperature, humidity, and cloud cover, acquired through cloud-integrated weather APIs and processed within the cloud analytics layer.

Fig. 14 illustrates the developed Node-RED-based solar power prediction dashboard for the next operational day. The time-resolved forecast, generated at three-hour intervals, enables operators to make informed scheduling decisions related to EV charging operations, Energy Storage System (ESS) charging, and grid interaction at the Point of Common Coupling (PCC).

The prediction results indicate a peak solar power generation of approximately **1.395 MW** during peak irradiance hours between **09:00 and 12:00**, while the average predicted generation is approximately **930 kW** over the effective solar window. Solar generation is comparatively lower during early morning and late evening hours, necessitating supplementary power support from either the ESS or the utility grid, depending on the available State of Charge (SOC).

This forecast is strategically aligned with historical demand patterns recorded in the system database. Grid peak demand typically occurs during mid-day fast-charging operations and evening battery swapping activities. By leveraging the forecasted solar availability, the PCC automation strategy prioritizes the following operational actions:

- Direct solar-to-load power delivery during peak irradiance hours,
- Opportunistic ESS charging during periods of surplus solar generation,
- ESS-assisted load support during grid peak demand intervals,
- Controlled grid power import during low solar availability and low SOC conditions.

This coordinated strategy enables effective load shifting from peak to off-peak periods, thereby reducing stress on the utility grid. Furthermore, it enhances renewable energy utilization by ensuring that solar energy is effectively exploited during peak demand intervals, contributing to improved grid stability and reduced power shortages.

VIII. RESULTS AND PERFORMANCE ANALYSIS

1) *Energy Management and System Performance*: The implemented IoT-enabled architecture demonstrated robust and effective energy management during hardware validation. A

D. Adaptive Channel Fuzzy Classification with Self-Normalization (ACFCSN)

The proposed Adaptive Channel Fuzzy Classification with Self-Normalization (ACFCSN) model is a lightweight, explainable, and context-aware anomaly detection framework designed for real-time IoT and Industrial IoT (IIoT) sensor communications. Unlike data-hungry supervised learning approaches, the ACFCSN model operates effectively on a limited number of channel observations, making it suitable for practical embedded and edge-based monitoring systems. The processing pipeline begins with multi-channel sensor data acquisition, where each channel corresponds to a monitored physical or communication parameter. To ensure robustness against incomplete data streams, mean-value imputation is applied to handle missing samples. This is followed by a self-normalization stage, which adaptively scales each channel based on its local statistical properties.

A Gaussian fuzzy membership function is employed to quantify the degree to which each normalized channel conforms to the learned *normal* system behavior. For a given channel value x_i , the fuzzy membership μ_i is computed as

$$\mu_i = \exp\left(-\frac{(x_i - \mu_i^*)^2}{2\sigma_i^2}\right), \quad (4)$$

where μ_i^* and σ_i denote the adaptive mean and standard deviation of the corresponding channel.

The individual channel memberships are aggregated to represent the overall system health. An adaptive decision threshold is defined as

$$T_{\text{adaptive}} = \bar{\mu} + k \cdot \sigma_{\mu}, \quad (5)$$

where $\bar{\mu}$ is the mean membership value, σ_{μ} is its standard deviation, and k is a tunable sensitivity parameter.

The system is classified as *NORMAL* if the aggregated membership exceeds the threshold; otherwise, it is classified as *ABNORMAL*, enabling fast and explainable anomaly detection.

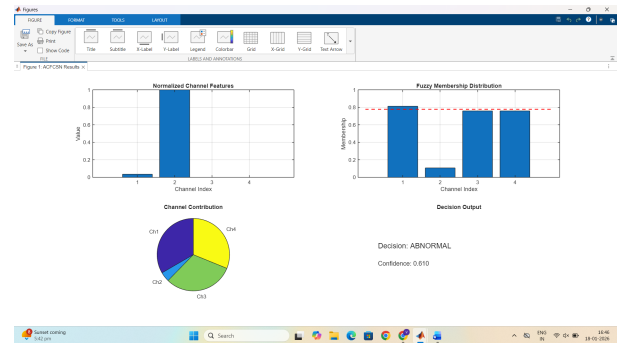


Fig. 15. ACFCSN model output illustrating normalized channel features, fuzzy membership distribution, relative channel contributions, and final anomaly classification decision.

renewable energy contribution ratio of 86.2% was achieved, indicating successful prioritization of solar energy utilization. Grid dependency was reduced to 11.5% of the total energy demand, with the remaining load supported by the Energy Storage System (ESS). The State of Charge (SOC) of the ESS was consistently maintained within the optimal operating

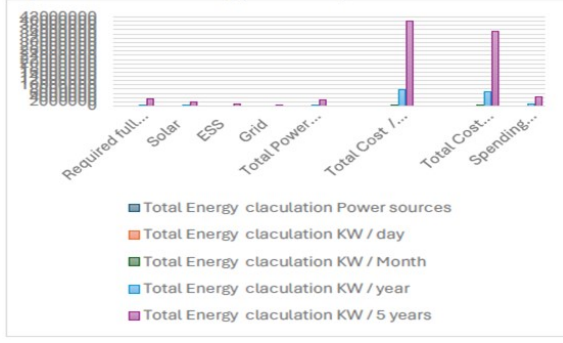


Fig. 16. Five-year energy contribution projections by source.

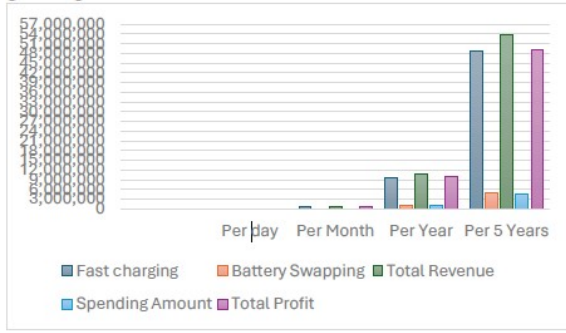


Fig. 17. Five-year cost and revenue projections.

range of 20–80%, validating the effectiveness of the proposed control and energy management algorithms. Transitions between solar, ESS, and grid operating modes were executed seamlessly with a measured switching latency below 200 ms, ensuring uninterrupted and stable power delivery to the connected EV charging loads.

2) *Long-Term Energy and Cost Projections:* Fig. 16 presents the projected energy estimation for the full-scale system over a five-year operational period, quantifying the sustainable energy contribution from renewable sources in comparison to grid dependency. The projection illustrates the cumulative impact of the proposed architecture, with renewable sources projected to supply approximately 2.93 million kWh over five years, accounting for 88.5% of the total energy demand.

The economic analysis, depicted in Fig. 16, demonstrates compelling financial viability. The projections show that while operational spending on grid power remains relatively flat, revenue from both fast charging and battery swapping services grows substantially over time. Based on these projections, the system is expected to achieve a cumulative profit of approximately Rs. 49.25 million over five years, corresponding to a cost-saving efficiency of 88.5% through optimized utilization of renewable energy sources.

A. Comparative Performance Metrics

Table I summarizes the key performance indicators comparing simulation predictions with experimental hardware measurements.

TABLE I
PERFORMANCE METRICS COMPARISON (SIMULATION VS. HARDWARE)

Metric	Simulation	Hardware
Renewable Energy Utilization	88.6%	86.2%
ESS Round-Trip Efficiency	92.0%	89.5%
DC Bus Voltage Regulation	$\pm 2.1\%$	$\pm 3.5\%$
PCC Transition Time	< 150 ms	< 200 ms

B. Economic Viability Analysis

The financial model, scaled to full-system deployment parameters, confirms the strong economic viability of the proposed EV charging infrastructure. A detailed daily operational analysis reveals the following performance indicators. The system generates approximately 1608 kWh of renewable energy per day, while grid dependency is limited to 209 kWh. The corresponding daily grid energy cost is approximately Rs. 2508, whereas the avoided grid cost due to renewable utilization amounts to Rs. 19,296 per day.

In addition, the combined revenue from DC fast charging and battery swapping services is estimated at Rs. 29,494 per day, resulting in a net daily profit of approximately Rs. 26,986. These results clearly demonstrate that the proposed hybrid energy management system not only enhances sustainability but also delivers substantial economic benefits under realistic operating conditions.

These figures translate to an annual profit of approximately Rs. 9.85 million for a single station configuration, establishing a compelling business case for deploying the proposed architecture at scale. The results conclusively demonstrate that the IoT-enabled hybrid energy management system not only achieves significant technical improvements in energy autonomy and grid stability but also delivers substantial economic benefits through optimized renewable integration and intelligent load management.

IX. CONCLUSION

This research successfully designed, simulated, and physically implemented a novel IoT-enabled hybrid energy management system for EV charging infrastructure, demonstrating the viability of a five-layer IoT architecture that intelligently integrates solar PV, Energy Storage Systems (ESS), and the grid via an automated Point of Common Coupling (PCC). The system's rule-based automation prioritized renewable sources, achieving a renewable energy contribution ratio of 86.2% and a cost-saving efficiency of 88.5% in hardware validation, which closely matched simulation projections. This performance translates to a significant reduction in grid dependency and a projected daily net profit of RS . 26,986 for a full-scale deployment. By moving from simulation to a functional hardware prototype, this work provides a validated, scalable blueprint for sustainable and economically viable EV charging networks, bridging a critical gap between theoretical models and real-world implementation for future smart grid integration.

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